

Kushagra Pandey

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Education

University of California, Irvine

PHD IN COMPUTER SCIENCE | 3.98/4.0

Irvine, California

Sep. 2022 - May, 2027 (Expected)

Indian Institute of Technology (IIT), Kanpur

MASTERS IN COMPUTER SCIENCE AND ENGINEERING | 10.0 / 10.0

Kanpur, India

Sept. 2020 - May, 2022

Indian Institute of Technology (IIT), Bhubaneswar

BACHELORS IN ELECTRICAL ENGINEERING | 8.87 / 10.0

Bhubaneswar, India

July, 2012 - May, 2016

Research Publications

Variational Test-time Optimization for Diffusion Synchronization

Hyunsoo Lee, Farrin Marouf Sofian, Kushagra Pandey, Stephan Mandt

PREPRINT, UNDER REVIEW

<https://arxiv.org/abs/2510.06637>

Hierarchical Variational Policies for Reward-Guided Diffusion

Kushagra Pandey, Farrin Marouf Sofian, Jan Niklas Groeneveld, Felix Draxler, Stephan Mandt

PREPRINT, UNDER REVIEW

<https://arxiv.org/abs/2510.06637>

Control-Augmented Autoregressive Diffusion for Data Assimilation

Prakhar Srivastava, Farrin Marouf Sofian, Francesco Immorlano, **Kushagra Pandey**, Stephan Mandt

UNDER REVIEW

<https://arxiv.org/abs/2510.06637>

On the Challenges and Opportunities in Generative AI

Laura Manduchi*, Clara Meister* **Kushagra Pandey***, et al.

TRANSACTIONS ON MACHINE LEARNING RESEARCH (TMLR)'25. SURVEY CERTIFICATION

<https://openreview.net/pdf?id=NeS9Kj2JwF>

Variational Control for Guidance in Diffusion Models

Kushagra Pandey*, Farrin Marouf Sofian*, Felix Draxler, Stephan Mandt

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML)'25

<https://arxiv.org/abs/2502.03686>

Heavy-Tailed Diffusion Models

Kushagra Pandey, Jaideep Pathak, Yilun Xu, Stephan Mandt, Mike Pritchard, Arash Vahdat, Morteza Mardani

INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS (ICLR)'25

<https://openreview.net/forum?id=toz10EN4qp>

Fast Samplers for Inverse Problems in Iterative Refinement Models

Kushagra Pandey*, Ruihan Yang*, Stephan Mandt

NEURAL INFORMATION PROCESSING SYSTEMS (NEURIPS)'24

<https://openreview.net/forum?id=qxS4IvtLdD>

Efficient Integrators for Diffusion Generative Models

Kushagra Pandey, Maja Rudolph, Stephan Mandt

INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS (ICLR)'24

<https://openreview.net/forum?id=qA4fox05Gf>

Towards Fast Stochastic Sampling in Diffusion Generative Models

Kushagra Pandey, Maja Rudolph, Stephan Mandt

NEURIPS 2023 WORKSHOP ON DIFFUSION MODELS

<https://arxiv.org/abs/2402.07211>

A Complete Recipe for Diffusion Generative Models

Kushagra Pandey, Stephan Mandt

INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV'23),

Oral Presentation (1.6% acceptance rate)

<https://arxiv.org/abs/2303.01748>

DiffuseVAE: Efficient, Controllable and High-Fidelity Generation from Low-Dimensional Latents

Kushagra Pandey, Avideep Mukherjee, Piyush Rai, Abhishek Kumar

TRANSACTIONS ON MACHINE LEARNING RESEARCH (TMLR)

<https://openreview.net/forum?id=ygoNPRiLxw>

Inference of cell state transitions and cell fate plasticity from single-cell with MARGARET

Kushagra Pandey, Hamim Zafar

NUCLEIC ACIDS RESEARCH (IF: 16.6)

<https://academic.oup.com/nar/article/50/15/e86/6593121>

VAEs meet Diffusion Models: Efficient and High-Fidelity Generation

Kushagra Pandey, Avideep Mukherjee, Piyush Rai, Abhishek Kumar

NEURIPS 2021 WORKSHOP ON DEEP GENERATIVE MODELS AND DOWNSTREAM APPLICATIONS

Oral Presentation (Top 5 of accepted papers)

https://openreview.net/forum?id=-J8dM4ed_92

Reviewer Experience

Served as a reviewer for top conferences/journals in ML including: ICML'26, ICLR'26, NeurIPS'25 (Top Reviewer Award), ICLR'25, TMLR

Broad Research Interests

Deep Generative Models (Diffusion Models/Flow Matching), Guidance and Efficient sampling in Generative Models

Research Experience

PhD Student

GRADUATE STUDENT RESEARCHER | SUPERVISOR: PROF. STEPHAN MANDT

UC Irvine

Sep 2022 - Present

- Exploring theoretical and practical aspects of continuous-time score-based generative models and related families.

Efficient, Controllable and High-Fidelity Generation from Low-Dimensional Latents

IIT Kanpur

MASTERS THESIS | SUPERVISOR: PROF. PIYUSH RAI

Jul 2021 - May 2022

- Worked on improving the sample quality of VAE's by hybrid generative modelling approaches for image synthesis.
- Worked on understanding the fundamental problems underlying the poor reconstruction quality of non-hierarchical or standard VAE's in general.

Elucidating cellular dynamics using Unsupervised Representation Learning in single-cell RNA-seq data

IIT Kanpur

RESEARCH ASSISTANT | SUPERVISOR: PROF. HAMIM ZAFAR

Dec 2020 - May 2022

- Working on developing Deep Latent Variable Models for multi-omic data integration
- Developed MARGARET: a deep unsupervised metric learning-based algorithm for trajectory inference in fundamental biological processes like cell differentiation using single-cell RNA-seq data.

Industry Experience

Long Horizon Modeling for Weather Forecasting

RESEARCH INTERN | SUPERVISOR(S): ARASH VAHDAT, MORTEZA MARDANI

- Worked on long-horizon forecasting on weather data.

[NVIDIA Research](#)

Jun 2025 - Current

Heavy-Tailed Diffusion Models

RESEARCH INTERN | SUPERVISOR(S): MIKE PRITCHARD, MORTEZA MARDANI, ARASH VAHDAT

- Worked on designing diffusion/flow matching models for modeling extreme weather events.

[NVIDIA Research](#)

Jun 2024 - Sep 2024

Efficient Samplers for Diffusion Generative Models

RESEARCH INTERN | SUPERVISOR: MAJA RUDOLPH

- Worked on improving the sampling efficiency of diffusion models by developing efficient frameworks amenable to numerical integration for diffusion model sampling.

[Bosch AI Research](#)

Jun 2023 - Sep 2023

Designing computational pipelines for genomics data

ML ENGINEER

- Worked on building the computing infrastructure (including DevOps and data management) for computational biology workflows for early prognosis of cancer.

[Lexent Bio Inc.](#)

Jun 2018 - May 2020

Academic achievements

- Recipient of the AI Residency at the Chan-Zuckerberg Initiative (CZI).
- HPI Fellowship Recipient at UCI.
- Received the Dean's Award at UCI for excellent research potential among incoming graduate students.
- Ranked 1st in a cohort of 100 students in the CSE department at IITK. Received the Academic Excellence Award for 2020-2021 and 2021-2022 for the same (Awarded to top 10% students).

Patents

Splitting integrators for fast sampling from diffusion generative models

PATENT APPLICATION: US20250103930A1

[Robert Bosch GmbH](#)

27th Mar 2025

Heavy tailed diffusion models

PATENT APPLICATION: PENDING

[NVIDIA Corporation](#)